

THARRANIAH TAMILWANAN

Data Science | Research & Data Analytics

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PROFESSIONAL SUMMARY

Final-year Bachelor of Computer Science (Hons.) Data Science student at Multimedia University, graduating July 2026, with hands-on research and data analysis experience at Merdeka Center for Opinion Research. Combines survey analytics, data cleaning, data validation, and sentiment analysis with Python-based machine learning and statistical analysis. Skilled at turning raw data into clear insights, reports, and visualizations. Seeking roles in data analytics, research analytics, business intelligence, and risk analytics.

EDUCATION

Multimedia University (MMU), Cyberjaya — BCS (Hons.) Data Science 2023 – 2026

CGPA: 3.63 | Dean's List: Term 2310, 2430 & 2530 | MUET Band 4.5

Multimedia University (MMU), Cyberjaya — Foundation in Information Technology Aug 2022 – Jul 2023

CGPA: 3.52

SMK (P) Sri Aman 2017 – 2022

SPM: 10A's | PT3: 9A's

WORK EXPERIENCE

Research & Data Analysis Intern — Merdeka Center for Opinion Research Jul 2025 – Oct 2025

- Compiled and organised election datasets across 17 Sabah constituencies to support research and accurate reporting.
- Assisted in preparing and analysing 10+ Monitoring & Evaluation (M&E) reports for parliamentary and community engagement workshops.
- Conducted data cleaning, validation, and statistical analysis in Google Sheets to support evidence-based reporting.
- Coordinated and documented multiple Focus Group Discussions (FGDs), contributing to qualitative analysis and stakeholder reporting.

FINAL YEAR PROJECT

A Comparative Study on Factuality in Large Language Models 2025 – Present

Python, TruthfulQA, LLM Evaluation, Statistical Analysis

- Designed a multi-metric evaluation framework to benchmark ChatGPT-4o mini, DeepSeek-V3.2, and Gemini 3 Flash across 817 questions in 38 categories (TruthfulQA dataset).
- Measured factual accuracy, hallucination rate, and label consistency: Gemini 3 Flash reached 82.86% factual accuracy and 7.53% hallucination rate; ChatGPT-4o mini led label consistency at 90.33%.
- Applied statistical comparison to assess LLM reliability, demonstrating the value of multi-metric frameworks for responsible-AI evaluation.

PROJECTS

Sales Performance Dashboard — Power BI 2025

Power BI, DAX, Data Modelling (Star Schema), Time Intelligence

- Built an interactive sales dashboard featuring revenue, profit, and order KPIs, month-over-month trend analysis, and product-category breakdowns on a star-schema model with custom DAX measures.

Bank Customer Subscription Predictor — ML Web App Nov 2025 – Feb 2026

Python, scikit-learn (Random Forest), Streamlit, UCI Bank Marketing Dataset (41,188 records)

- Built an end-to-end predictive analytics web app classifying term-deposit subscription with probability scores, risk levels, and tailored recommendations for non-technical users.

Handwritten Digit Recognition System Nov 2025 – Jan 2026

- Built a multi-digit recognition pipeline (Python, MNIST, classical computer vision) and deployed it as a Streamlit app for real-time inference.

SKILLS

Programming & Databases: Python, R, SQL, C++

Data Analysis & ML: Pandas, NumPy, scikit-learn, Random Forest, Classification, Predictive Analytics, Statistical Analysis

Research & Analytics: Survey Data Cleaning, Data Validation, Sentiment Analysis, Quantitative Analysis, Trend Analysis, Insights Generation, Research Methodology

Visualization & Reporting: Power BI (Basic), Matplotlib, Excel, Google Sheets, Data Visualization, Report Writing, Canva

Tools: Jupyter Notebook, Streamlit, Microsoft Office, LaTeX

Languages: English (Fluent), Malay (Fluent), Tamil (Native)

LEADERSHIP & ACTIVITIES

Integrity & Leadership Day 2023 (Registration) | **Blood Donation Campaign 2025** (Logistics) | **BarCamp 2025** (Event Management)

Treasurer, Computer Club (SMK) – Managed club finances | **St. John Ambulance** – Certified First Aider